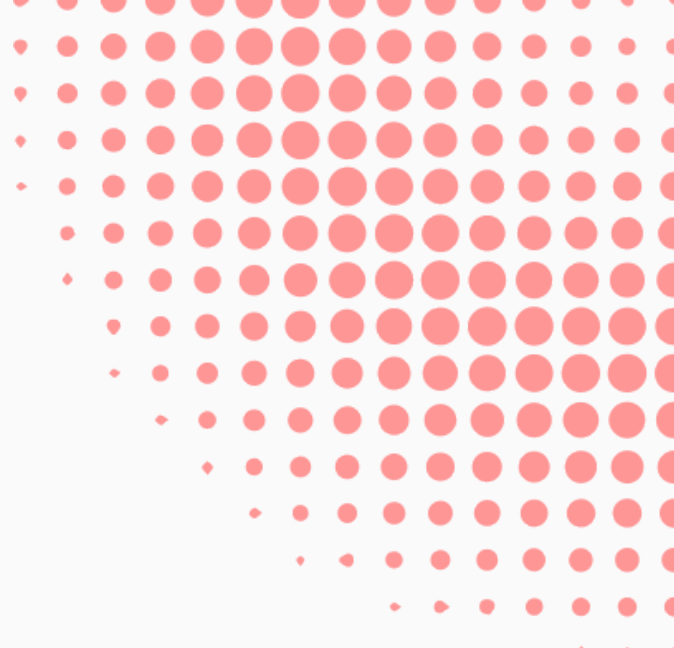




ANALYSIS OF HUMAN REACTION TO DIFFERENT MUSCLE STIFFNESS AND APPLICATION TO THE ROBOT

by Ben Srisoontorn

日本のプレゼンテーション



What is the study about?

Pharse 1: Analysing the relationship between muscle stiffness and therapist's reaction

- Setting the scope of study
- Identifying required data and existing resource
- Designing the data collecting methodology
- Collecting Data
- Analysing Data

Pharse 2: Integating the result to make better response for robot

- Designing control system to the robot to resemble the result from analysis
- Integate the control system to the robot

Pharse 2 work may not be done due to the restriction of time during internship period

This week's progress

CURRENT STAGE

1

**Clarify
the scope
of the
research**

2

**Identify required
data for the
analysis and
existing resource**

3

**Design
data
collecting
process**

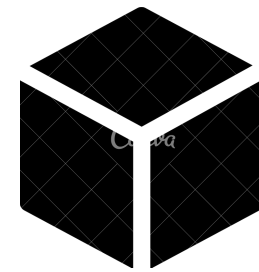


**FUTURE
WORK**

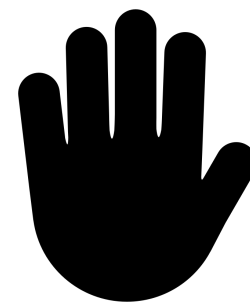
Scope of research



The reaction of therapist to **Muscle stiffness only** not to bone



Only on **flat surface**



Contact area on **forehand only**

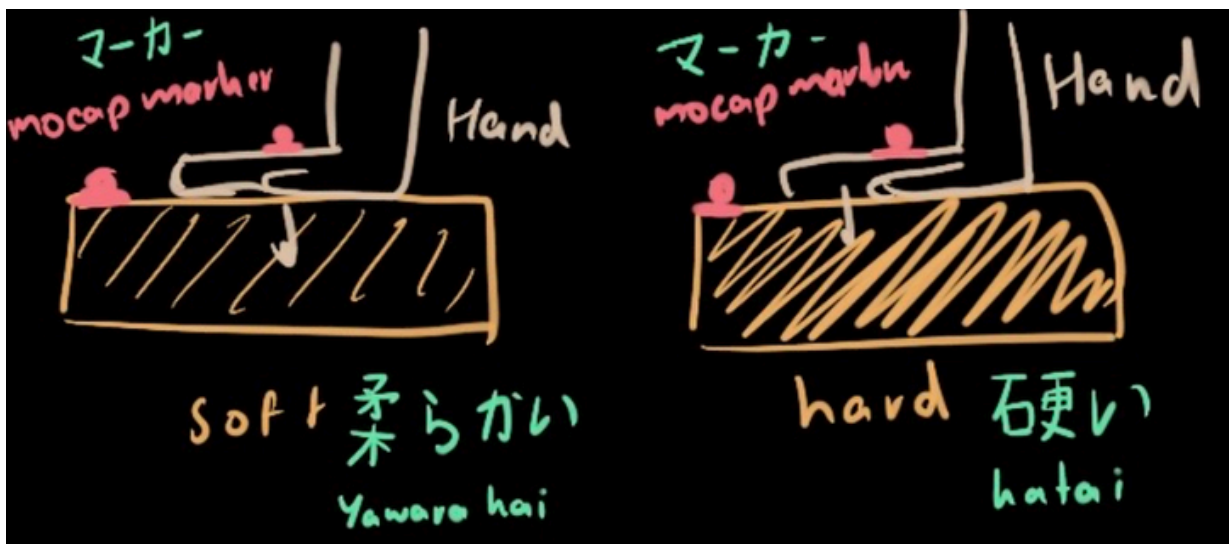
Required Data

		Method to acquire data
1	Movement of Hand	Motion Capture Glove
2	Speed of Hand Movement	Motion Capture Camera
3	Position of Hand	Motion Capture Camera
4	Force of the Hand and Contact Surface	Force sensitive sensor
5	Timing of Each Movement	Recording Timestamp

Experiment Design

1

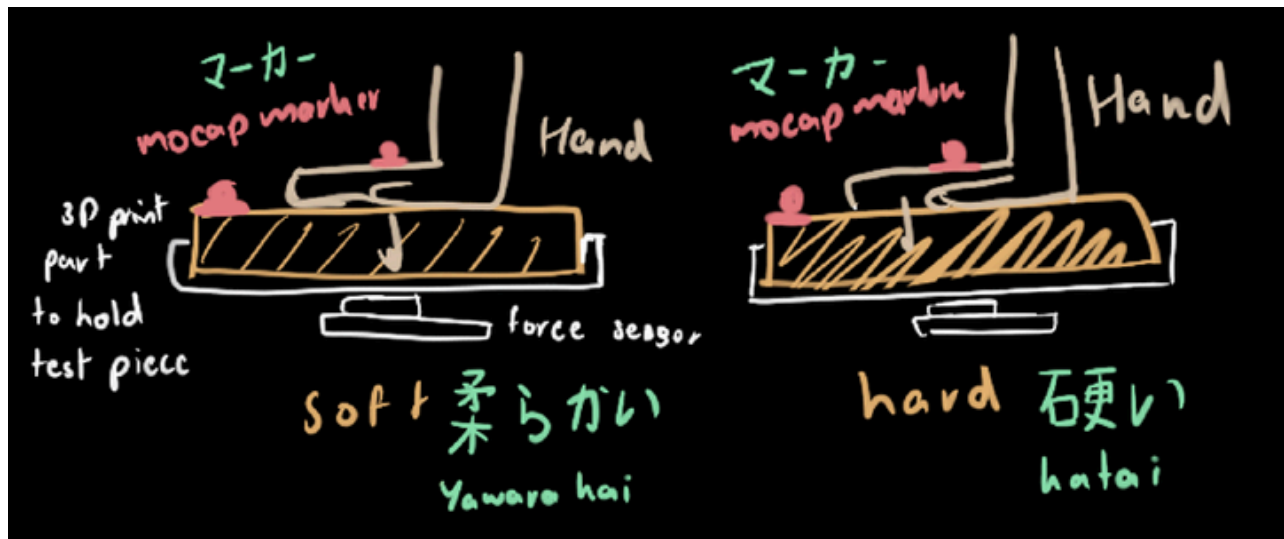
Force sensitive resistor on glove



Need permission to purchase sensor and other components

2

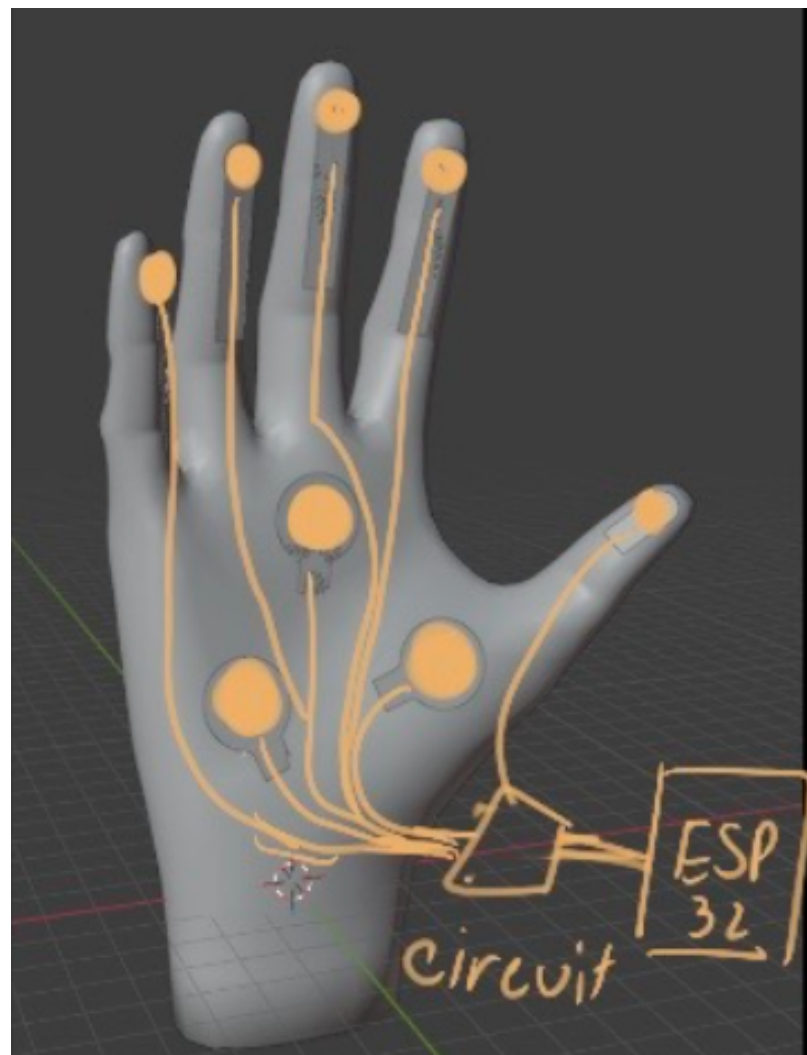
Force sensor from under test piece



Can be done now (no detailed plan)

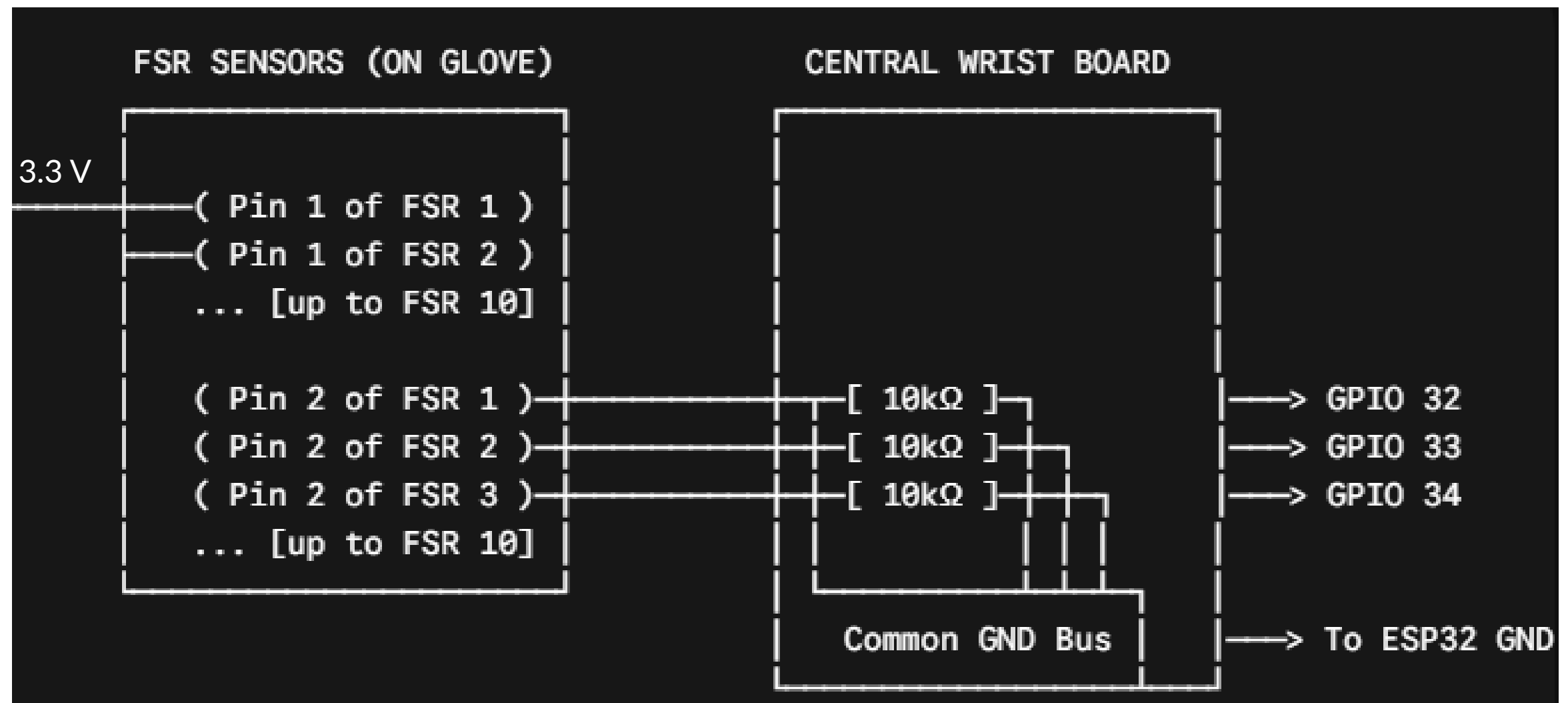
Force sensitive resistors on glove

Position on glove



3 on palm, 5 on each tip of finger

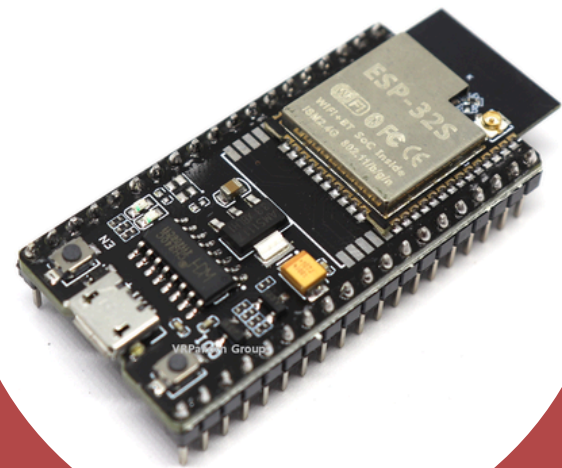
Circuit



use ESP 32 as microcontroller to recieve data and sent to computer through WIFI to keep in form of CSV file for further analysis

Force sensitive resistors on glove

Component



**ESP32 and
Upload Cable**



FSR

- FSR400 long X4
- FSR400 short X1
- FSR402 short X3



10K resistor



Wires

Future Work

- Get the premission to make **FSR glove**
- Design the experiment to find **the force applied to different hardness of material from under test pieces**
- Analyse the measured result
- Research **reference paper** to clarify the analysis